Brief Introduction to Quantum Algorithms

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Visit to the University of Tokyo

I am honored to visit the Department of Physics at the University of Tokyo.

I am especially grateful to **Professor Mio Murao** for this opportunity.

If you have questions about quantum computing, I'd be delighted to chat during my stay (until November 21st).

Feel free to talk to me when I'm at the university or contact me via email: `Pawel.Wocjan@ibm.com`.

Context and Focus

With steady progress in quantum hardware, error correction, and fault tolerance,

the search for new quantum algorithms for large-scale fault-tolerant quantum computers has become more critical than ever.

Challenges in Design and Analysis of Quantum Algorithms

Sweet Spot: Find problems with:

- Practical relevance
- Classical hardness
- Quantum efficiency

Provable quantum advantage: Rigorously prove separation between classical and quantum algorithms.

Heuristic quantum advantage: Test heuristics on large fault-tolerant quantum computers.

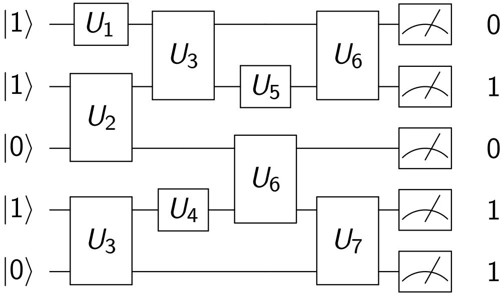
Some Quantum Subroutines and Algorithms

- Quantum phase estimation & Quantum Fourier transform
- Block encodings, quantum signal processing
- Simulation of quantum systems
  - Hamiltonian time evolutions
  - Lindbladian time evolutions
  - Preparing thermal states (Gibbs sampling)
  - Preparing ground states
- Amplitude amplification and estimation
  - Grover's algorithm
  - Quantum Chebyshev inequality
- Quantum walk-based algorithms

Some Quantum Subroutines and Algorithms

- Quantum algorithms for
  - Linear equations
  - Semidefinite programming
  - Differential equations
- Quantum algorithms for the hidden subgroup problem
  - Shor's algorithms for integer factorization & computing DLOG
  - Algorithms for computing unit & class groups of number fields
- Quantum algorithms for optimization
  - Adiabatic algorithm
  - Decoded quantum interferometry
- Quantum algorithms for problems in topology
  - Data analysis
  - Khovanov homology, Jones and HOMFLYPT polynomials

Quantum Circuit Model



Quantum circuits: Foundational model for fault-tolerant quantum computing.

Qiskit: Open-source framework for designing, simulating, and running quantum circuits.

`https://www.ibm.com/quantum/qiskit`

Quantum Phase Estimation

Given: A unitary U and eigenvector $|\psi\rangle$ with unknown eigenphase $\varphi \in [0, 1)$ where

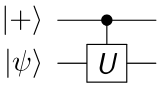
$$U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle$$

Task: Output an estimate $\hat{\varphi}$ such that

$$\Pr\left(|\hat{\varphi} - \varphi| \leq \frac{1}{2^n}\right) \geq \frac{3}{4}$$

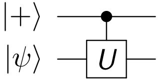
Deviation $|\hat{\varphi} - \varphi|$ is computed modulo 1.

Phase Kick Back: Derivation



$$
\begin{aligned}
\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |\psi\rangle \quad &= \quad \frac{|0\rangle}{\sqrt{2}} \otimes |\psi\rangle + \frac{|1\rangle}{\sqrt{2}} \otimes |\psi\rangle \\
&\mapsto \quad \frac{|0\rangle}{\sqrt{2}} \otimes |\psi\rangle + \frac{|1\rangle}{\sqrt{2}} \otimes U|\psi\rangle \\
&= \quad \frac{|0\rangle}{\sqrt{2}} \otimes |\psi\rangle + \frac{|1\rangle}{\sqrt{2}} \otimes e^{2\pi i \varphi}|\psi\rangle \\
&= \quad \frac{|0\rangle}{\sqrt{2}} \otimes |\psi\rangle + \frac{e^{2\pi i \varphi}|1\rangle}{\sqrt{2}} \otimes |\psi\rangle \\
&= \quad \frac{|0\rangle + e^{2\pi i \varphi}|1\rangle}{\sqrt{2}} \otimes |\psi\rangle
\end{aligned}
$$

Phase Kick Back: Summary



$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |\psi\rangle \mapsto \frac{|0\rangle + e^{2\pi i\varphi}|1\rangle}{\sqrt{2}} \otimes |\psi\rangle$$

The eigenstate $|\psi\rangle$ in the target register remains unchanged.

$\Rightarrow$ Focus on the control register.

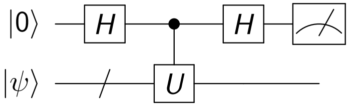
The state $|0\rangle + |1\rangle$ of the control qubit changes to $|0\rangle + e^{2\pi i\varphi}|1\rangle$ via phase kick back.

Hadamard Test + Phase Kick Back

The Hadamard transformation H is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The Hadamard test circuit is

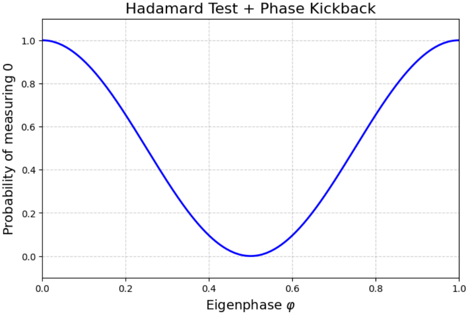


The second Hadamard gate transforms the state as follows:

$$\begin{aligned}
& \frac{|0\rangle + e^{2\pi i \varphi}|1\rangle}{\sqrt{2}} \\
\mapsto \quad & \frac{1}{2}\big((|0\rangle + |1\rangle) + e^{2\pi i \varphi}(|0\rangle - |1\rangle)\big) \\
= \quad & \frac{1}{2}\big((1 + e^{2\pi i \varphi})|0\rangle + (1 - e^{2\pi i \varphi})|1\rangle)\big) := |\varphi\rangle
\end{aligned}$$

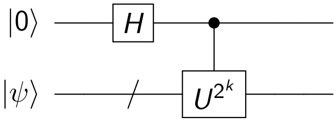
Hadamard Test + Phase Kick Back

$$
\begin{aligned}
\Pr(0) &= \left\| |0\rangle\langle 0| \, |\varphi\rangle \right\|^2 \\
&= \left| \frac{1}{2} \left(1 + e^{2\pi i \varphi}\right) \right|^2 \\
&= \frac{1}{2} \left(1 + \cos(2\pi\varphi)\right)
\end{aligned}
$$



Phase Kick Back Using Powers of U

For arbitrary k, we obtain



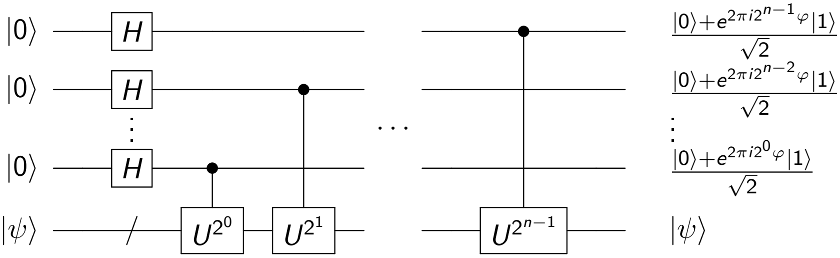
$$\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 2^k \varphi}|1\rangle)$$

$$|\psi\rangle$$

since

$$U^{2^k}|\psi\rangle = e^{2\pi i 2^k \varphi}|\psi\rangle$$

Phase Kick Back Part of Phase Estimation



We set

$$|\varphi\rangle := \frac{|0\rangle + e^{2\pi i 2^{n-1}\varphi}|1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + e^{2\pi i 2^{n-2}\varphi}|1\rangle}{\sqrt{2}} \otimes \cdots \otimes \frac{|0\rangle + e^{2\pi i 2^{0}\varphi}|1\rangle}{\sqrt{2}}$$

Binary Fractions

Assume that the eigenphase φ is an exact n-bit binary fraction, i.e.,

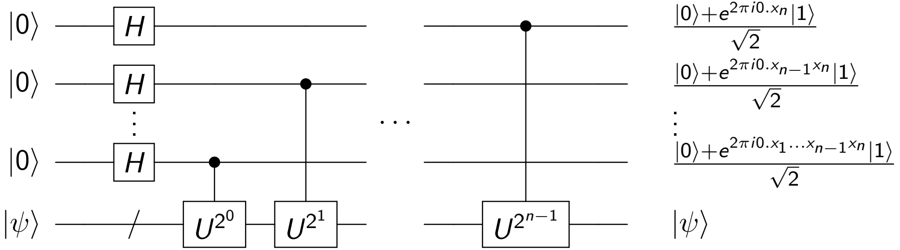
$$\varphi = 0.x_1x_2\ldots x_n = \sum_{i=1}^{n} \frac{x_i}{2^i}$$

For $k \in \{0, \ldots, n-1\}$, we have

$$2^k\,\varphi = x_1x_2\ldots x_k.x_{k+1}\ldots x_n$$

$$\begin{aligned}
e^{2\pi i 2^k \varphi} &= e^{2\pi i(x_1x_2\ldots x_k.x_{k+1}\ldots x_n)} \\
&= e^{2\pi i(x_1x_2\ldots x_k+0.x_{k+1}\ldots x_n)} \\
&= e^{2\pi i(x_1x_2\ldots x_k)} \cdot e^{2\pi i(0.x_{k+1}\ldots x_n)} \\
&= e^{2\pi i(0.x_{k+1}\ldots x_n)}
\end{aligned}$$

Phase Kick Back Part of Phase Estimation: Summary



Quantum Fourier Transform

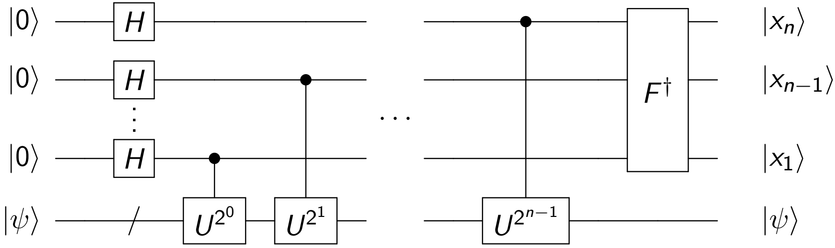
The QFT F is defined by

$$F\big(|x_n\rangle \otimes |x_{n-1}\rangle \otimes \cdots \otimes |x_1\rangle\big)$$

$$= \frac{|0\rangle + e^{2\pi i 0.x_n}|1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle}{\sqrt{2}} \otimes \cdots \otimes \frac{|0\rangle + e^{2\pi i 0.x_1 x_2 \ldots x_n}|1\rangle}{\sqrt{2}}$$

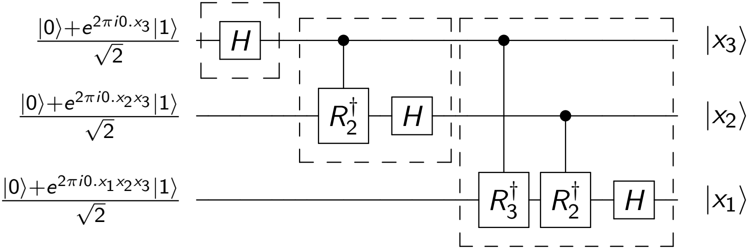
$\Longrightarrow$ We can use inverse QFT $F^\dagger$ to obtain the bits of the eigenphase.

Quantum Circuit for Phase Estimation



Let us now derive the quantum circuit for $F^\dagger$.

Inverse QFT for 3 bits

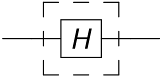


The phase shift R_k is defined by

$$R_k := \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix}$$

Least Significant Bit – Top Qubit

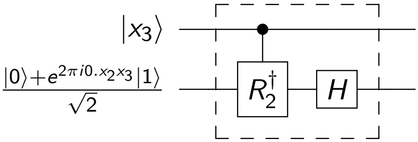
$\frac{|0\rangle + e^{2\pi i 0.x_3}|1\rangle}{\sqrt{2}}$



$$\frac{|0\rangle + e^{2\pi i 0.x_3}|1\rangle}{\sqrt{2}} = \frac{|0\rangle + (-1)^{x_3}|1\rangle}{\sqrt{2}}$$

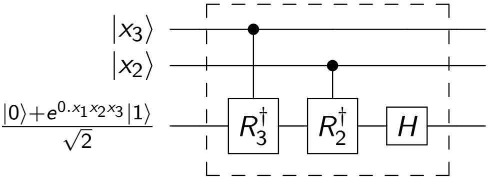
$$\xrightarrow{H} \quad |x_3\rangle$$

Second Bit – Middle Qubit



$$
\begin{aligned}
& |x_3\rangle \otimes \frac{|0\rangle + e^{2\pi i 0.x_2 x_3}|1\rangle}{\sqrt{2}} \\
\xrightarrow{ctrl\ R_2^\dagger} \quad & |x_3\rangle \otimes \frac{|0\rangle + e^{2\pi i 0.x_2 0}|1\rangle}{\sqrt{2}} \\
\xrightarrow{I \otimes H} \quad & |x_3\rangle \otimes |x_2\rangle
\end{aligned}
$$

Most Significant Bit – Bottom Qubit



$$|x_3\rangle \otimes |x_2\rangle \otimes \frac{|0\rangle + e^{2\pi i 0.x_1x_2x_3}|1\rangle}{\sqrt{2}}$$

$$\xrightarrow{ctrl\ R_3^\dagger} \quad |x_3\rangle \otimes |x_2\rangle \otimes \frac{|0\rangle + e^{2\pi i 0.x_1x_2 0}|1\rangle}{\sqrt{2}}$$

$$\xrightarrow{ctrl\ R_2^\dagger} \quad |x_3\rangle \otimes |x_2\rangle \otimes \frac{|0\rangle + e^{2\pi i 0.x_1 00}|1\rangle}{\sqrt{2}}$$

$$\xrightarrow{I \otimes I \otimes H} \quad |x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle$$

Summary of Phase Estimation Circuit

Phase kick back using the controlled U^{2^k} gate prepares the state

$$\frac{|0\rangle + e^{2\pi i 0.x_{k+1}x_{k+2}\ldots x_n}|1\rangle}{\sqrt{2}}$$

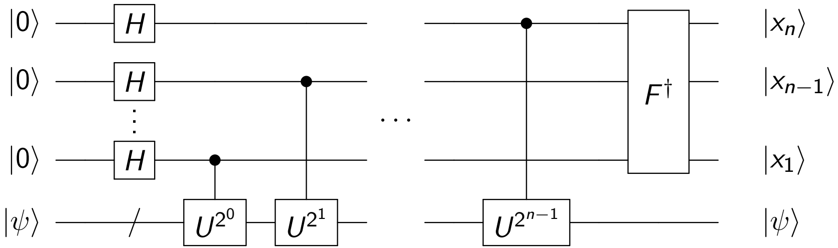
Phase shifts controlled by the previously determined bits $x_{k+2}, \ldots, x_n$ transforms it to

$$\frac{|0\rangle + e^{2\pi i 0.x_{k+1}0\ldots 0}|1\rangle}{\sqrt{2}} = \frac{|0\rangle + (-1)^{x_{k+1}}|1\rangle}{\sqrt{2}}$$

The Hadamard gate transforms it to

$$|x_{k+1}\rangle$$

Summary of Phase Estimation Circuit



The measurement of the qubits yields the bits $x_n, x_{n-1}, \ldots, x_1$ deterministically.

General Case: Arbitrary Eigenphases

If φ is not an exact n-bit fraction, the inverse QFT

$$F^\dagger|\varphi\rangle$$

yields a superposition of n-bit strings.

Measuring the qubits results in a probabilistic outcome.

Probability of Measuring a Bit String

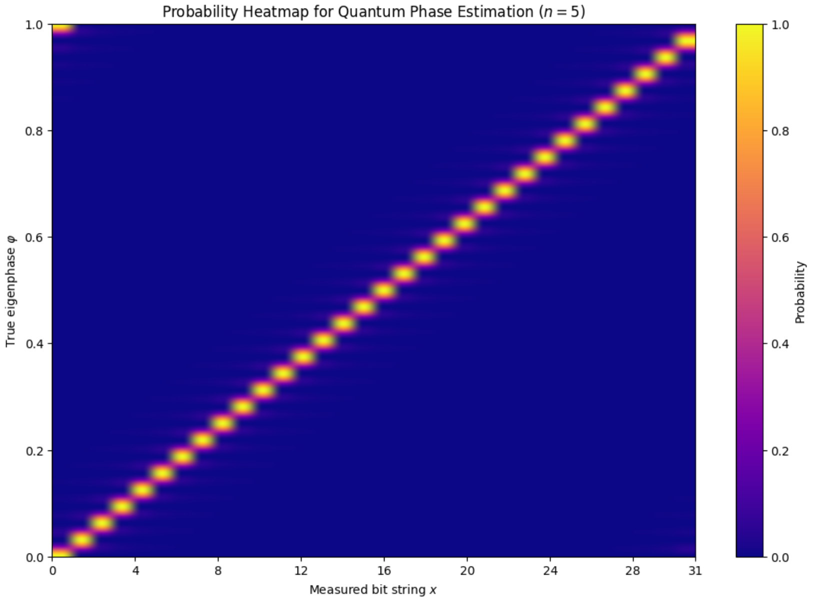
Let $x = \sum_{k=1}^{n} x_k 2^{n-k}$ and $\varphi_x = 0.x_1 x_2 \ldots x_n = \frac{x}{2^n}$ be the corresponding n-bit fraction.

The probability of measuring φ_x is:

$$\Pr(x) = \frac{1}{2^{2n}} \frac{\sin^2\left(2^n \pi(\varphi - \varphi_x)\right)}{\sin^2\left(\pi(\varphi - \varphi_x)\right)}$$

The probability of measuring either of the two closest n-bit fractions to φ is at least $\frac{3}{4}$.

Quantum Phase Estimation: Probability Heatmap



Physics Applications of Quantum Phase Estimation

Quantum methods can efficiently simulate the time evolution:

$$U_t = \exp(-iHt)$$

for a large class of physical Hamiltonians (e.g., local Hamiltonians).

Applying QPE to U_t enables:

- Resolving the energies of H (idealized).
- Implementing projectors onto low-energy states, which – combined with amplitude amplification – can prepare quantum states supported on low energies.
- Implementing jump operators $S(\omega)$ for Bohr frequencies ω, facilitating the simulation of a Davies generator (Lindbladian) to prepare thermal states.

Note: Perfect energy resolution is not feasible in practice.

Shor's Algorithm: Quantum Phase Estimation Approach

Reduction:

Factoring

$\Downarrow$

Order finding

$\Downarrow$

Phase estimation of a unitary

Order Finding

Given $a, N \in \mathbb{Z}$ with $\gcd(a, N) = 1$, the **order** of a modulo N is the smallest positive integer r such that $a^r \equiv 1 \pmod{N}$.

- Given: $a, N \in \mathbb{Z}$ with $\gcd(a, N) = 1$
- Task: find the order r of a modulo N

Reduction of Factoring to Order Finding

Choose $a \in \{2, 3, \ldots, N-1\}$ uniformly at random.

If $\gcd(a, N) \neq 1$, then it is a nontrivial factor of N.

If $\gcd(a, N) = 1$, let r denote the order of a modulo N.

Output $\gcd(a^{r/2} - 1, N)$ as a non-trivial factor of N.

This succeeds with large probability.

Example Reduction for Small N

Consider $N = 15$ and $a = 7$:

$$7^1 = 7, \quad 7^2 \equiv 4, \quad 7^3 \equiv 13, \quad 7^4 \equiv 1 \quad \implies \quad r = 4$$

We have

$$\gcd(a^{r/2} - 1, N) = \gcd(7^2 - 1, 15) = \gcd(48, 15) = 3$$

Order Finding via Quantum Phase Estimation

Order finding can be understood as applying quantum phase estimation to the unitary:

$$U|x\rangle = |ax \pmod N\rangle,$$

where $x \in \{0, 1, \ldots, N-1\}$.

When restricted to the invariant subspace $|1\rangle, |a\rangle, \ldots, |a^{r-1}\rangle$, U is isomorphic to a cyclic shift modulo r.

The eigenphases of the cyclic shift operator are fractions:

$$\frac{k}{r}, \quad k \in \{0, \ldots, r-1\}.$$

The denominator r is the order.

We obtain samples close to these fractions and recover the order r.

Shor's Algorithm as a Hidden Subgroup Problem over $\mathbb{Z}$

Define the function:

$$f : \mathbb{Z} \to \mathbb{Z}_N, \quad x \mapsto a^x \pmod{N}.$$

This function is:

- Periodic with period r
- Injective within each period

The symmetry is characterized by the **hidden subgroup** $H = r\mathbb{Z}$.

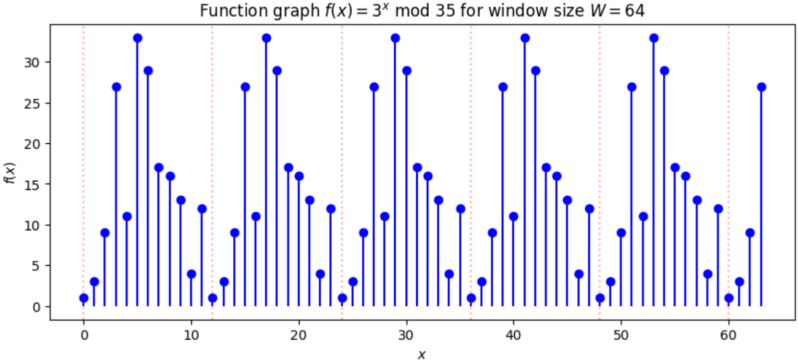
Function Evaluation in Input Superposition

- Since $\mathbb{Z}$ is infinite, we use a finite window $[0, \ldots, W-1]$ with $W = 2^n$ on a quantum computer.
- Evaluate f in superposition over the window:

$$\frac{1}{\sqrt{W}} \sum_{x=0}^{W-1} |x\rangle \otimes |f(x)\rangle.$$

- This quantum state encodes the function graph of f.

Example Function Graph



Collapse onto Preimage After Output Measurement

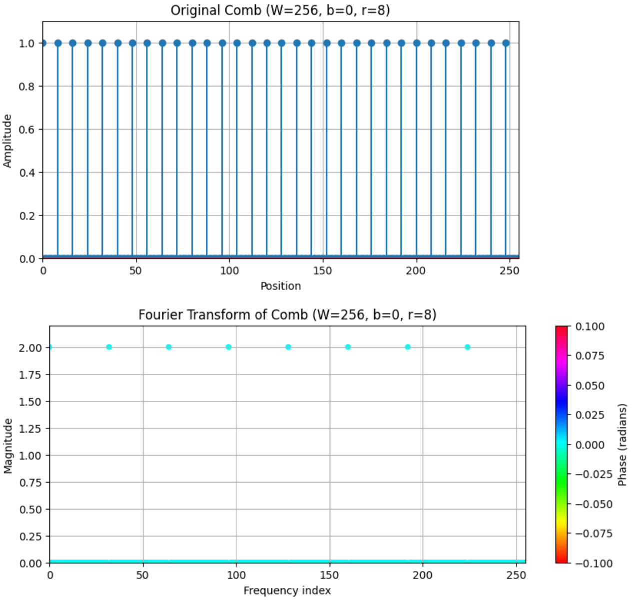
Let $y \in \mathrm{Im}(f)$. The preimage of y is the set:

$$f^{-1}(y) = \{x \in \mathbb{Z}_W : f(x) = y\} = \{b + \ell \cdot r\},$$

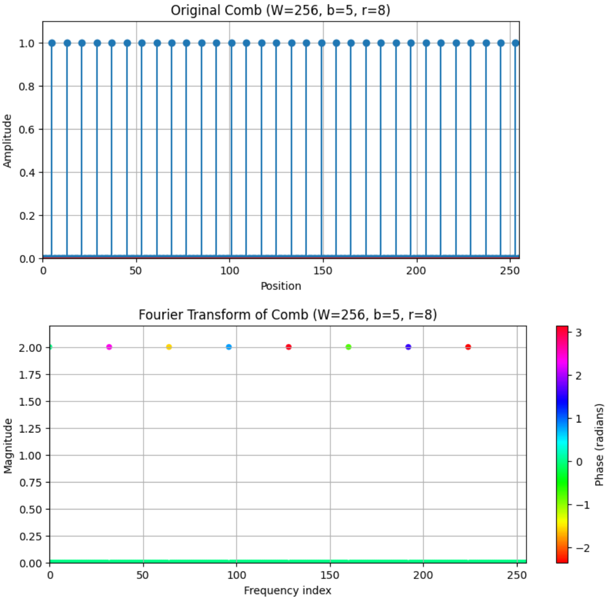
where b is the smallest value in the window with $f(b) = y$ and ℓ ranges over all values such that $b + \ell r < W$.

Measuring the output register collapses the input register into a **translated comb state** with offset b and period r.

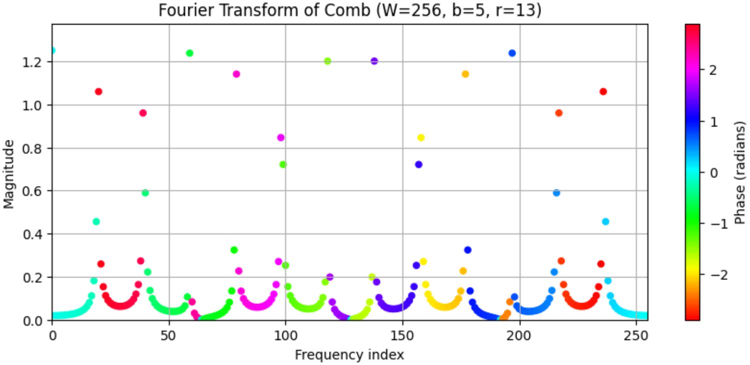
Perfect Periodicity, No Offset



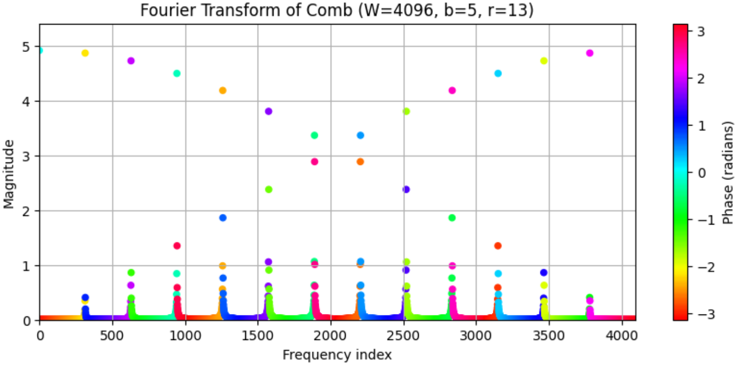
Perfect Periodicity, Offset



Imperfect Periodicity



Imperfect Periodicity, Larger Window



Hidden Subgroup Problem

- Let G be a group and H a subgroup.
- Let $f : G \to S$ (where S is some set) be a function such that

$$f(g) = f(g') \iff gH = g'H$$

  In words, two group elements g and g' have the same image under f if and only if they are in the same coset in G/H.
- Task: given an oracle for f, determine the hidden subgroup H.

Hidden Subgroup Problem: Algorithms and Applications

- Efficient quantum algorithms exist for finite abelian and some finite non-abelian groups.
- Efficient quantum algorithms also exist for the infinite abelian groups $(\mathbb{Z}, +)$ and $(\mathbb{R}^n, +)$.
- Graph isomorphism can be cast the HSP over the symmetric group S_n.
- Shor's algorithm for factoring solves the HSP over $\mathbb{Z}$.
- Shor's algorithm for the DLOG problem in $\mathbb{F}_q^*$ solves the HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ for $N = q - 1$.

Quantum Algorithm for Abelian HSP: Preparing a Random Coset State

- We prepare the function graph state:

$$\frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle \otimes |f(g)\rangle$$

- Measuring the output register collapses the input register onto a random coset state:

$$|t + H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |t + h\rangle,$$

where $t \in T$ and T is a transversal for the coset space G/H.

QAlgo for Abelian HSP: Pontragyagin Duality

- Pontragyagin duality establishes an isomorphism φ between a finite abelian group G and its character group $\hat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$.
- This isomorphism allows us to **identify** G with $\hat{G}$, mapping each element $g \in G$ to a character $\chi_g \in \hat{G}$.
- For a subgroup $H \leq G$, the dual subgroup $H^\perp$ is defined as:

$$H^\perp = \{g \in G : \chi_g(h) = 1 \quad \forall h \in H\}.$$

In words, $H^\perp$ is the collection of characters than annihilate each element in H.

QAlgo for Abelian HSP: Fourier Transform and Recovery

- The quantum Fourier transform F_G over G is given by:

$$F_G = \frac{1}{\sqrt{|G|}} \sum_{g,g' \in G} \chi_g(g') |g'\rangle\langle g|$$

- It maps the uniform superposition over H to the uniform superposition over $H^\perp$:

$$F_G|H\rangle = |H^\perp\rangle$$

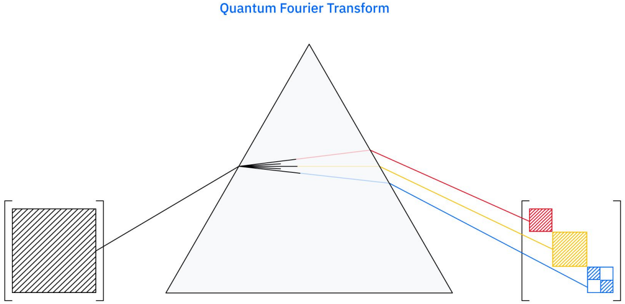
- Using an oracle for f, we prepare a random coset state:

$$|t + H\rangle$$

- Applying F_G to $|t + H\rangle$ yields a phase-modulated $|H^\perp\rangle$.
- We can sample uniformly at random from $H^\perp$ and recover the hidden subgroup H.

Non-Abelian Fourier Transforms: A Bird's-Eye View

- The **quantum Fourier transform for a group** G decomposes **coset states** into components corresponding to the **irreducible representations** of G.



- For **Abelian groups**, these representations are one-dimensional.

Non-Abelian Fourier Transforms: Recent Progress by IBM

- While QFT has been successful for **Abelian groups** , its extension to **non-Abelian groups** has historically faced challenges.
- Recent work by IBM researchers has demonstrated a **quantum algorithm for non-Abelian Fourier transforms**, specifically for the symmetric group.

  `https://www.ibm.com/quantum/blog/group-theory`
- This algorithm provides a **potential quantum speedup** for computing Kronecker coefficients and other multiplicities, with implications for algebraic combinatorics and quantum advantage.

Resources for Quantum Algorithms

- **Book:** *An Introduction to Quantum Computing* by Phillip Kaye, Raymond Laflamme, and Michele Mosca
- **Lecture Notes:** *Lecture Notes on Quantum Algorithms* by Andrew Childs: `https://www.cs.umd.edu/~amchilds/qa/`
- **Recent Developments:** For more recent topics such as
  - *block encoding & quantum signal transform*
  - *decoded quantum interferometry*
  - *quantum Gibbs sampling*

  it is best to consult original research articles, arXiv preprints, or tutorials on platforms like YouTube.